

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-abe86d4bf17a8700e.jsonl`  
- SHA-256: `05c02298d61bf2d27f160e633167bff4d77117546f7378eab73dc6303274dbd5`  
- Source modified: `2026-07-02T02:59:39+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:58:05.140Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.
A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra (avidullu/khelsutra):

```
{  
  "title": "Docs claim the retired droplet alpha is LIVE (droplet deleted 2026-06-25)",  
  "category": "doc-rot",  
  "severity": "P1",  
  "file": "NEXT_STEPS.md",  
  "line": 12,  
  "evidence": "NEXT_STEPS.md:12-13 ('Where we are (updated 2026-07-01)') still says the site is  
'live + browsable on the DO droplet -> http://168.144.126.176:8787'; deploy/DEPLOY_ALPHA.md:3  
says 'Status: LIVE (deploy live-verified on the droplet 2026-06-22)' and :29 'This runbook ships  
the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE  
(2026-06-22)'. Project memory (MEMORY.md:63, session-handoff.md:19): droplet claude-do-rally-  
test/168.144.126.176 RETIRED+deleted 2026-06-25; alpha paused pending Cloud Run. nslookup shows  
api/app.khelsutra.guru still resolve via Cloudflare to a tunnel with no origin (pending  
CNAME/Access cleanup).",  
  "proposed_fix": "One doc PR: mark DEPLOY_ALPHA.md status RETIRED/PAUSED with a pointer to the  
Cloud Run plan, fix NEXT_STEPS.md 'Where we are' (droplet gone, site is CF-Workers-only), and  
correct LOCAL_APPLIANCE_ARCHITECTURE.md:176. Also note the single-origin repoint that superseded  
the split topology (issue #97). The 2026-07-01 refresh (#139) touched docs but kept the stale  
claim, so this will keep misleading ramp-up reads.",  
  "effort": "S",  
  "tracked_in": "partially in issue #97 (item 2: update DEPLOY_ALPHA.md); the NEXT_STEPS.md  
droplet claim is untracked"  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer,
C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-
data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator,
C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field;
non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup',
C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).
VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited
code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent
commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete
failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it
owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training,
product/strategy/licensing, real cloud spend, counsel ToS/DPA?
HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only
commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you
personally saw the defect/debt in current code; if you cannot reproduce the claim from the code,

say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:58:15.187Z)

```
1      # Next steps (live) ? KhelSutra brand / site
2
3      Prioritized "what to do next" for the `khelsutra` product repo1. Refresh at each
checkpoint.
4
5      **Where we are (updated 2026-07-01):** `khelsutra.guru` is **LIVE and fully shipped2 ?
landing page + `/capture`
6      checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7      text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8      D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9      live). **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to
`master` ? deploys;
10     **CI gates every PR** ? three jobs: the full site `npm test` (Worker + VM host + assets
+ UX), the
11     `web/` SPA build + coverage floor, and a real-browser layout check. The same form also
runs on a
12     dependency-free **Node + SQLite** host (`site/server.js`), **live + browsable on the DO
droplet ?
13     http://168.144.126.176:8787** (systemd `khelsutra-site`). Shipped across ~90 merged PRs
(the form,
14     6-language localization, the per-rally MVP, droplet portability). Docs:
15     `docs/REQUEST_ACCESS_FORM.md`, `docs/PORTABILITY.md`.
16
17     ## Next product threads (owner ideas, 2026-06-20)
18     - **? DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22; human gates cleared):** the
product is
19     non-tech-friendly and **fully localized to all 6 languages** (en/kn/hi/es/da/id)
across both surfaces
20     (`web/` SPA + `site/` marketing) ? including the home page (hero, all sections, the
`#join` access form,
21     footer), the **Privacy Policy** page, **per-page ``s**</b>, and <b>**es/da/id academy</b>
<b>flyers**</b> (with
22     localized SPA screenshots). The responsive header is fixed and a <b>**headless-Chromium</b>
<b>CI suite**</b> guards
23     layout, cross-locale render (live DOM == catalog), and <b>**accessibility (axe-core WCAG</b>
<b>2.1 AA)**</b>. Full
24     record: <b>**`docs/archives/FRIENDLY_AND_MULTILINGUAL.md`**</b>. Post-archive PRs: #86 (own-
model) · #87 (es/da/id
25     site) · #105 (form+footer + render guard) · #106 (Privacy page) · #107 (a11y fixes +
axe guard + page
26     titles) · #109 (flyers). <b>**Owner gates CLEARED:**</b> native-speaker review of the AI-
drafted catalogs ? +
27     real-device font-glyph eyeball ?. <b>**Remaining (small/owner):**</b>
28     - <b>**Delta native-review**</b> of the strings ADDED after the owner's sign-off
(<b>#105/#106/#107/#109 ?</b> the
29     form/footer/privacy/title/flyer copy is AI-drafted): a small top-up review before a
wide public launch.
30     - <b>**B7 (deferred, not a v1 blocker):**</b> engine backend customer-facing l10n (report
<b>`customer_`</b> +
31     rejection reasons) ? cross-repo Python (<b>`badminton-highlight-indexer`</b>), owner-gated
<b>**D-BACKEND-DEPTH**</b>.
32     - <b>**Marketing video on the site**</b> ? embed a short walkthrough of the record ? index ?
reel process,
33     themed for <b>**serious enthusiasts**</b>, on the landing page. ? <b>**A real-UI CUJ screencast</b>
<b>now exists**</b>
34     (Playwright recording of the per-rally SPA driven against real GPU-run data ?
<b>`promo_cuj.mp4/`.gif`</b>
35     in <b>`gs://khelsutra-rally-corpus/promo/`</b>). Decide: use it as the marketing clip (?</pre></div>
```

```

shows real court
36      footage + a dev "P5" badge ? clean up before public embed), and/or produce a polished
Grok/Gemini cut.
37      - **Interactive per-rally selection** (big product feature) ? ? **MVP / M0 COMPLETE**
(the full
38      pick ? query ? select ? compile ? download loop on existing engine endpoints). Lives
in the alpha SPA
39      (`web/`). **? Tracked design + P-tracker (source of truth):
`docs/PER_RALLY_SELECTION.md` (IN PROGRESS;
40      owner-approved, D8?D11 locked). **Done: P0 design (#11), P1 scaffold (#14), P2
grid+hook (#21), P3
41      `parseQuery()` query bar (#28), P4 SelectionFooter + `min_shot_count` warning (#30),
P5 real load +
42      Build?compile?preview?download (#31). ** Verified e2e vs a stub engine + a CUJ replay
against the real
43      L4-run rallies + reel. **NEXT (owner-gated, engine repo): M1?M5** ? `GET /api/health`
+ `download_url`
44      (P6), RESTful `~/rallies` alias (P7), multi-video home + auth (P8), per-rally
preview/timeline (P9),
45      corpus correction loop (P10), localized NL (M5). **Optional:** a fresh live
`/api/process` GPU
46      inference once the engine session's M2 training frees the single GPU.
47
48      ## Queued (site polish)
49      - **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
50      - **Cloudflare Web Analytics (owner-gated; FREE)** ? code side DONE (the dead
placeholder beacon was
51      removed from `site/public/index.html`; analytics is now via **Automatic Setup**, no
in-source token).
52      **Owner step (needs CF login):** the site is a Cloudflare **Worker** (not Pages) on
the proxied
53      `kheldutra.guru` zone ? enable via **Analytics & Logs ? Web Analytics ? Add a site ?
`kheldutra.guru`**
54      (Automatic Setup is default-on for proxied hostnames; edge-injects the beacon on every
page). Then
55      filter the view to `hostname = khelsutra.guru` to exclude `*.pages.dev`/preview noise,
and decide
56      separately whether `app.kheldutra.guru` (the SPA) needs its own enablement (history-
router only).
57      **Verify (two gotchas):** the Worker's asset responses must NOT send `Cache-Control:
?, no-transform`
58      (Cloudflare can't rewrite that ? no injection), and after a deploy the rendered HTML
should contain the
59      edge-injected beacon. **DONE WHEN:** the Web Analytics dashboard shows incrementing
page views for
60      `kheldutra.guru`.

```

3. user (2026-07-02T02:58:16.131Z)

```

1      # Deploy KhelSutra Studio for alpha ? a shareable page, no SSH
2
3      > **Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22 ?
`deploy/droplet/README.md`; only a rotated Gemini key + a browser OTP login remain to exercise
the upload?reel CUJ) . **Topology:** split ? SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel, Cloudflare Access over both (the engine's `HOSTING_PLAN.md` $2 alpha plan).
. **Cost:** ~$0/mo standing (Pages + Tunnel + Access are free ?50 users) + Gemini per-run
(~$0.05/match).
4
5      This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a
**page you can share with alpha testers** ? they open `app.kheldutra.guru`, sign in (email
allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.
6
7      > **Outgrown the droplet?** To move the live engine onto a **GCP L4 GPU VM** (more
CPU/RAM for ffmpeg
8      > + optional WASB GPU detection) and cut the tunnel over to it, see

```

```

**[GPU_VM_SERVING.md](GPU_VM_SERVING.md)**
9      > ? same tunnel + Access, no DNS change, droplet kept as fallback (~$500?600/mo while
the VM runs).
10
11      ## 0. The shape
12
13      ```
14      tester browser
15      ? https
16      ?
17      Cloudflare edge (free): DNS · TLS · Access (email-OTP allowlist) in front of BOTH
hostnames
18      ??? app.khelsutra.guru ? Cloudflare Pages          (the web/ SPA; built with
VITE_API_BASE)
19      ??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on
YOUR box
20
21
22
23
24
25      - **No inbound ports.** `cloudflared` dials out; the engine stays bound to
`127.0.0.1:8000`. The box's home/public IP is never exposed.
26      - **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also*
verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a
direct hit to the tunnel can't spoof identity.
27      - **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ?
the SPA is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine
locks CORS to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's
already env-driven).
28
29      > **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA
*and* the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
`LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
(stable Pages-hosted page + a thin API tunnel).
30
31      ---
32
33      ## 1. Prerequisites
34
35      - A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU;
fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed;
the `badminton-highlight-indexer` engine checked out.
36      - A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing
site there).
37      - `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
38      - A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks
to Google; this is a paid + cloud action).
39
40      ---
41
42      ## 2. Engine (the `api.` side)
43
44      ### 2.1 Install (with the auth extra)
45      ```bash
46      cd badminton-highlight-indexer
47      pip install -e .[auth]          # PyJWT[crypto] ? REQUIRED when edge auth is on, else
every authenticated request is rejected 401
48      ```
49
50      ### 2.2 The safe-exposed config
51      Create / edit `config.json` on the box. **Where it must live:** the engine reads

```

`\$RALLY\_APP\_HOME/config.json` if `RALLY\_APP\_HOME` is set, else `./config.json` relative to *the directory you launch the server from* ? **not** necessarily the repo root. If the engine reads a different (or absent) file it silently falls back to all-defaults, which means **edge auth is OFF**. To make writer and reader agree by construction, write it with `python -m backend.main --setup` (it targets the exact path the server reads) and edit *that* file:

```
52     ``jsonc
53     {
54         "security": {
55             "edge_auth_enabled": true,                // turn the CF-Access JWT verify ON
56             "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when
exposed
57             "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER
allowed_video_root (see §7)
58             "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
59             "cf_access_aud": "<the CF Access application AUD>" // filled in §4
60         },
```

4. user (2026-07-02T02:58:16.997Z)

```
160     | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
161     | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ?
|
162     | T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `/api/upload/`
? server path ? `/api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
163     | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
164     | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
165     | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner ? **engine half
shipped on `origin/master`** (`/api/capabilities` main.py:340, `/api/health` :307); only the SPA
banner remains | T1 | coord | ? | engine (on master) |
166     | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
167     | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
168     | T10 | Observability pass: correlation id end-to-end + structured logs + metrics (§5) |
T4 | no | ? | ? |
169     | T11 | Retention sweep for `output/`+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges
stale uploads >7d + orphan `.partial-` + regenerable frames/staged/proxies; **reels/DB/CSVs
durable** ; `--json` plan ready for a scheduler. Automated scheduling ? [engine
#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ?
(uploads 7d, keep reels ? 2026-06-21) | ? | [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |
170     | T12 | Ingest from **bucket URI** (`s3:///`gs://`) wiring ? engine accepts URIs
(M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])** ; in-UI `gs:///`reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` §9) | T4 | ? | ? | engine #280 + SPA #56 |
171     | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native` ; owner
byte-parity (P4) | branch | ? owner | ? | ? |
172     | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
173
174     **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does **not** block on it (ships on Gemini).
175
176     ** ? Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).**
The alpha ships the **split** topology (SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel + CF Access over both ? the engine `HOSTING_PLAN.md` §2), **not** the single-
origin Caddy shell of **T2** (kept as the documented alternative ? §0/D1). The **deploy kit +
runbook** ([`deploy/DEPLOY_ALPHA.md`](../deploy/DEPLOY_ALPHA.md) +
`deploy/cloudflared.config.example.yml`) is shipped (? templates+runbook). **The alpha stand-up
is now LIVE (2026-06-22):** the engine booted on the droplet (`168.144.126.176`),
`app.khelsutra.guru` is up, and the full **DNS ? Cloudflare ? Access ? tunnel ? engine** chain
```

plus the **T3** raw-:8000-refused posture test (boot-posture fail-fast, Host-guard `400?200`, path-jail, T4 upload?jail, edge-auth `401-gate`) were all **exercised live** **[deploy/droplet/README.md:100-106]**. Only the **Gemini-key rotation** + a **browser OTP login** remain. Enabled by: engine exposure-safety boot posture (engine #294) + SPA **VITE_API_BASE** ([#64]).

177

178 **## 10. Open questions ? owner / external**

179 **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling [#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:** auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates with the active worker sessions.

180 **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming endpoint; per-user scoping timing; containerization vs runbook.

181

182 **## 11. Definition of done**

183 - [~] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via the ingest menu. (2026-06-22: chain LIVE end-to-end ? DNS?Access?tunnel?engine, T4 upload?jail exercised; the full upload?reel pass still needs the Gemini key + a browser OTP login **[deploy/droplet/README.md:100-106]**.)

184 - [x] Raw `:8000/`:900x` unreachable from outside (**tested** live 2026-06-22, not assumed) **[deploy/droplet/README.md:100-106]**.

185 - [] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped honestly (not faked).

186 - [] RallyPicker embedded with honesty rules; **min_shot_count** surfaced.

187 - [] **/api/capabilities** + **/api/health** + banner truthful; reload-survival via **/api/videos** + **/api/reels**.

188 - [] **Observability:** correlation id threaded end-to-end; structured logs at every new path; loud failures; logs inspected during CUJ replay.

189 - [] Retention sweep in place (**tool shipped** [engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-schedule** [engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **GEMINI_API_KEY** rotated; **allowed_video_root** set.

5. user (2026-07-02T02:58:27.932Z)

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\compute-stack-gcp-only.md:16:- The \$200 DO credits, if used, go ONLY to non-compute DO bills cleanly (CPU droplet / site host / Spaces storage / bandwidth) ? **never GPU**.

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\compute-stack-gcp-only.md:17:- **deploy/digitalocean/** scripts (**bootstrap.sh**/**mount_scratch.sh**/**gpu_setup.sh**) are KEPT ? they're **provider-agnostic** Linux-GPU setup reused by the GCP runbook; only DO **provisioning** (doctl, Spaces) is retired. Optional later: move them to **deploy/common/**.

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\compute-stack-gcp-only.md:19:- The shared DO droplet **claude-do-rally-test** (other session's testbed) + retiring the DO API token = owner-coordinated, separate from this. See [[gcp-vm-always-delete]] [[gcp-ops-agent-enable]] [[monetization-plan]].

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:18:[Omitted long matching line]

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:19:[Omitted long matching line]

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:20:- **STATE:** alpha LIVE on the droplet (CPU, **gemini** segmenter) at **api.khelsutra.guru**; GPU-VM-serving is documented not deployed; 0 of my PRs open; GCP \$0 (all validation resources torn down); only sibling worktrees remain. [[cloudrun-gpu-serving-gotchas]] [[gotcha-shared-checkout-branch-collisions]]

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:22:**Checkpoint:** 2026-06-23 (**CLOUD TOGGLE LIT + VALIDATION METHOD + LIVE E2E PROVEN + user guide + droplet full-suite**). master `688d28f`. Owner: "light up the cloud compute toggle with Cloud Run + a good validation method to ensure the intended path was taken" ? then

```

"run the test on the droplet with a full install" ? "write a doc to help users set up + create
reels." ALL DONE:
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:27:- **? Droplet FULL-install suite GREEN:** pushed master?`root@168.144.126.176:~/bhi-
fulltest`, fresh venv + `pip install -e .[dev,auth,storage-gcs,storage-s3]` + torch CPU +
ffmpeg/cv2 libs ? **1461 passed, 12 skipped** (the skips = the private `obsreport` git+ssh dep,
not installed). Droplet restored (dir removed, 1.6G freed).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:29:- **? NEXT:** merge #343+#344 on green; clean worktrees (bhi-computeval/bhi-
userdoc/bhi-verifyfix). Owner follow-up to make the toggle live on the ALPHA = deploy a
persistent Cloud Run Job + flip the droplet engine to cloud-serving (a deploy decision, not
code). [[cloudrun-gpu-serving-gotchass]] [[gcp-vm-always-delete]] [[working-agreement-merging]]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:31:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:49:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:51:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:54:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:73:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:78:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:79:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:82:- **RAMP-UP KIT:** this block ? the 13 issues above ?
`khelsutra/deploy/DEPLOY_ALPHA.md` + `deploy/droplet/` ? [[next-session-ux-localization]].
**Owner relationship: warm, high-trust, delegates ("take control"), fast real-world feedback,
strong product instincts.** [[gotcha-shared-checkout-branch-collisions]] [[working-agreement-
merging]] [[feedback-launch-quality-bar]] [[compute-stack-gcp-only]]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:84:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:85:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:87:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:88:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:99:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:101:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:103:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:104:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:108:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:116: - **Site (excite-to-join) [#34/#35]:** after sign-up the reveal now shows **(1)
the tool in action** (self-hosted footage-free `tool-cuj.mp4`) **+ (2) a ~30s reel preview**
(`REEL_SECS` 90?30, gated nocookie YouTube ? real faces stay in the unlisted reel, not the
repo); `revealAfterSignup`; coherent CTA/teaser copy; `server.js` serves `.mp4` as `video/mp4`;
`.mp4`/*.gif` marked binary. Site suite **34 green**; **droplet re-synced + parity GREEN**
(index+mp4 byte-identical CF?VM:8787).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:122:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:125:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:126:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-
state.md:127:[Omitted long matching line]

```

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:133:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:136:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:144:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:148:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\current-state.md:161:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\gcp-vm-always-delete.md:20:**Also billing (not GCP, not auto-deletable):** the DO droplet `claude-do-rally-test` (bl1r1, CPU `s-2vcpu-4gb`, ? \$24/mo) is a **shared testbed** (another session's engine + the site twin) ? owner-gated cleanup, not mine to delete. [[gcp-ops-agent-enable]]
[[monetization-plan]]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\golden-corpus-grew-9-to-15.md:14:(9 multicourt + 6 single-court). Earlier eval docs/tests that say the original-6 are "retired / labels
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:24:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:26:**Both-hosts compatibility now ENFORCED in CI (PR #7):**
`khelsutra/.github/workflows/test.yml` runs `npm test` on every push/PR (Node 24) ? **17 green** (12 Cloudflare-logic in `src/index.test.js` + 5 VM-host in `src/server.test.js`, which boots the real `server.js`). CI confirmed green on master. Live VM check: `SSH\_TARGET=root@host site/scripts/droplet-smoke.sh` (scp + boot + round-trip; PASSED on the droplet). So any change that breaks either host fails CI.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:28:[Omitted long matching line]
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:30:**SESSION WRAP (2026-06-20, part 2 ? PRs #9?#26, all merged; master `9f0942b`):**
Work continued well past launch; site is byte-in-sync across **Cloudflare + the DO droplet twin** (`/srv/khelsutra-site`).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:34:- **Provider-portable hosting** (#16): `site/scripts/provision-vm.sh` + `teardown-vm.sh` make destroy?resume-on-any-fresh-VM a tested, scripted, first-class flow (LIVE-verified on a throwaway DO droplet ? operational ? state-export ? destroy); `git -c core.autocrlf=false archive` CRLF fix (#19) keeps Windows-operator syncs LF. Runbook = `docs/PORTABILITY.md`.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-domain-launch.md:37:- **Deploy mechanism:** site changes auto-deploy to Cloudflare (git-connected Worker on merge) AND are re-synced to the droplet via `git -c core.autocrlf=false archive HEAD:site | ssh ? tar -x -C /srv/khelsutra-site/site` + `systemctl restart khelsutra-site`, with `scripts/parity-check.sh` confirming byte-parity.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:2:name: khelsutra-site-droplet-parity
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:3:description: RETIRED 2026-06-25 ? the khelsutra DO droplet (site mirror + alpha engine) was deleted; data archived to GCS, parity monitor removed (PR #117). History + re-sync recipe kept as reference.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:10:**?? STATUS (2026-06-25): the droplet is DELETED.** Owner chose to retire it. The DO droplet
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:14:1.7 GB was preserved: reels + `sports\_indexer.db` + `access.db` + the 3 non-corpus uploads ? **`gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`** (+ local `C:/Users/avidu/Backups/claude-do-rally-test-20260625/`); the 2 real match videos (`GX010142/143\_1080p.mp4`) were already in `~/videos/`. `doctl compute droplet delete` done; no orphan volumes/IPs/snapshots ? \$0. Monitor removed via **khelsutra PR #117**.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:16:need cleanup/repoint (alpha ? Cloud Run M7); droplet runbook docs (`deploy/droplet/README.md`,
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-droplet-parity.md:28:- **`http://168.144.126.176:8787` (DigitalOcean droplet `claude-do-rally-test`)** ? a **MANUAL-sync
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-

```

droplet-parity.md:32:**The droplet drifts (? nightly parity failure) every time a site PR
lands**, because CF deploys
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:33:automatically but the droplet does not. First seen 2026-06-23/24/25: the
2026-06-22 localization PRs
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:35:but not the droplet, so it served the pre-i18n site. **Diagnosis rule: CF
== git HEAD, so the droplet
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:38:**Re-sync recipe** (verified 2026-06-25; key = `~/.ssh/khelsutra_droplet`,
the default ~/.ssh key is
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:39:NOT authorized; host alias `root@168.144.126.176`):
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:45:ssh -i ~/.ssh/khelsutra_droplet root@168.144.126.176 'tar czf
/srv/khelsutra-site/site_bak_$(date +%s).tgz -C /srv/khelsutra-site site'
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:46:git -C "$CLONE" -c core.autocrlf=false archive origin/master:site | ssh -i
~/.ssh/khelsutra_droplet root@168.144.126.176 'tar -x -C /srv/khelsutra-site/site'
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:47:ssh -i ~/.ssh/khelsutra_droplet root@168.144.126.176 'systemctl restart
khelsutra-site.service'
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\khelsutra-site-
droplet-parity.md:52:deploy step with the droplet key as a GH secret). The droplet is deleted
(see STATUS above); CF is the
c:\Users\avidu\.claude\projects\c--Users-avidu-Projects-khelsutra-guru\memory\MEMORY.md:32:-
[GCP VM: always DELETE, never just stop](gcp-vm-always-delete.md) ? owner cost rule
(2026-06-20): a STOPPED VM still bills its disk (~$34/mo for 200GB pd-ssd); **DELETE** ? $0
(boot disk autoDelete=true). `teardown.sh` now defaults to `delete`; audit `gcloud compute disks
list` for orphans. DO droplet `claude-do-rally-test` (~$24/mo) is a shared testbed (owner-
gated).

```

[Showing results with pagination = limit: 60]

6. user (2026-07-02T02:58:29.149Z)

```

9223ba9 docs: refresh khelsutra docs for today's storage-track work (P2-P5) (#139)
dd0394d docs: tracked project ? Studio media lifecycle (ingest?store?annotate?train) (#120)
6e4dbaf docs(deploy): GPU_VM_SERVING.md ? move the live engine to an L4 GPU VM off the droplet
(#113)
bc4e7ba docs: close out the Friendly & Multilingual chapter ? gates cleared, localization 100%
(#110)
ba38f15 fix(site): remove dead analytics beacon; use Cloudflare Automatic Setup (#100)
7dc7de5 docs: freshness sweep ? reconcile 13 docs with the live repo (+ engine origin/master)
(#95)
50b2bd7 chore(repo): ignore local runtime artifacts + correct stale NEXT_STEPS claims (#94)
e2570e4 docs: archive the Friendly & Multilingual tracker (engineering complete) (#93)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
bea8a2a docs(appliance): flip T11 retention sweep to ? ? tool shipped (engine #299), scheduling
tracked (#301) (#69)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
a0475bc docs: hygiene refresh ? sync trackers to shipped reality (#59)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
f9eea14 web: resilient video-summary readers + show fps in the library (#140)
9223ba9 docs: refresh khelsutra docs for today's storage-track work (P2-P5) (#139)
fb2f561 web: persistent "FFmpeg not detected" banner (hard-blocker warning) (#138)

```

7. user (2026-07-02T02:58:45.954Z)

```

1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? badminton-highlight-indexer (rename PENDING) + siblings: sports-
data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-25 (**review-automation + audit-completion session** ? see ?
You are here (2026-06-25) below. Maintenance/PR-review phase, NOT serving/alpha: 0 open PRs, all
CI green, no GCP VMs, PR-review watch loop OFF. The serving/alpha track below is UNCHANGED +
owner-gated.) Prior: 2026-06-22 (**M11+M12 serving-code session** ? owner: "build 3 then 4 then
we pair on the alpha (2)". **Both Claude-doable engine items DONE + merged:** **M11** data-
flywheel skeleton [#336](https://github.com/Khelsutra/badminton-highlight-indexer/pull/336)
(`334ad0f` ? default-OFF + consent FAKED-DENY, inert; reuses workspace/eval/promotion; NO
migration) + **M12** `gdrive-api://` OAuth StorageBackend
[#338](https://github.com/Khelsutra/badminton-highlight-indexer/pull/338) (`819e395` ?
idempotent put, lazy Google imports, injected-fake tests; live per-user OAuth owner-gated). Both
adversarially reviewed, full-suite+cross-OS green, merged on green, in an isolated worktree. **?
NEXT = pair with owner on the ALPHA GO-LIVE (option 2)** ? owner brings CF dashboard + box +
rotated Gemini key; the counsel ToS/DPA inputs (the 7-pt checklist) gate M11 *live* +
M9-consent. ? schema ladder drifted under me this session: v7=`locale`, v8=compute-picker ? the
future M9-consent migration is **v9** (check `PRAGMA user_version` before authoring). Prior
same-day: **maintenance / green-PR merge-sweep session** ? owner asked to merge the green PRs +
finish loose ends. 6 PRs now on master: worker dated-label fix **319**, nightly Cloud-Run
scheduler **320**, DATA_IN_GCS map **316**, doc **321**, gate-A litmus re-ground **322**,
golden-fixtures backlog **323**. engine `origin/master` = `9b37269`, suite **1428?**, **0 open
engine PRs**; docs-archive had no terminal candidate (owner-gated). Full per-PR detail =
[[current-state]] top. ? ~10 concurrent worktrees/sessions ? owner is trimming concurrency
before new work. The serving/alpha plan below is UNCHANGED. Prior **2026-06-21 ALPHA-SHAREABLE
session** ? Claude's half of "make it UI-driven + alpha-shareable" is **DONE**: 3 PRs merged ?
engine **294** exposure-safety boot posture + khelsutra **64** SPA `VITE_API_BASE` + **65**
deploy kit. **KEY CORRECTION:** B-P2 (SPA ingest screen) was ALREADY shipped (#56) ? alpha-
shareable was a DEPLOY/AUTH task, not an SPA build. Owner chose the **SPLIT** topology (SPA on
CF Pages + `api.*` cloudflared tunnel + CF Access). Remaining = OWNER stand-up via
`khelsutra/deploy/DEPLOY_ALPHA.md`; then engine M11/M12. Full record [[ui-driven-cloud-serving-
gap]]). Prior: HUGE serving session ? M1?M9-auth shipped (#279?#291) + real M7 L4 run. **Repo is
under the `Khelsutra` GitHub org**.
13
14     > **Read order:** this file ? [[current-state]] top (dense per-checkpoint detail) ?
`docs/NEXT_STEPS.md` (the prioritized cross-track plan, refreshed 2026-06-21) ?
`docs/DOC_STATUS.md` (doc lifecycle/hygiene).
15
16     ## ? You are here (2026-06-25)
17     **Review-automation + audit-completion session.** engine `origin/master` = **0b4f492**
. all CI green . **0 open PRs** . no GCP VMs . **PR-review watch loop OFF** (cancelled). NOT
serving/alpha (that track is the "Earlier" block below, unchanged + owner-gated).
18     - **Repo tidy:** synced master, pruned ~92 stale local branches + 7 stale worktrees;
**fixed a corrupted (UTF-16-tail) `.gitignore` ? `artifacts/` ignored +
`KHELSUTRA_PRODUCT_OVERVIEW.md` tracked (**377**); owner un-hid the hidden
`C:\Users\avidu\Projects` folder.
19     - **khelsutra site-droplet RETIRED** ([[khelsutra-site-droplet-parity]]): the dual-
purpose `claude-do-rally-test`/`168.144.126.176` (marketing-site mirror + alpha engine) kept
drifting from CF ? **deleted ($0; data archived to `gs://khelsutra-rally-
corpus/archive/droplet-?-20260625/` + local), parity monitor removed (khelsutra #117), Node-20
actions bumped (#118).** ? This means the prior "LIVE alpha on the droplet" in the Earlier block
is **GONE** ? alpha is paused pending the Cloud-Run move; `api.khelsutra.guru` CNAME/CF-Access
need cleanup.
20     - **CODE_CLEANUP_AUDIT advanced ? 3 HIGH items + infra all merged:** B2/P17 surface-

```

Gemini-failures (**#378**) · 01/P11 Gemini remote-file GC (**#380**) · B1/P9 remove-fabricated-motion-data (**#385**); + ruff-format split (#379), audit-doc reconcile/accuracy (#375/#376), Tier-1 nightly `nightly.yml` (#381), per-job CI timeouts (#382). **Remaining audit items: D1 (split `main.py` ~1450 lines), D13 (ByteTrack?`trackers`), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4). **

21 - **NEW tracked project: Decision-Snapshot Regression** ([[decision-snapshot-regression]] · `docs/DECISION\_SNAPSHOT\_REGRESSION.md` · **#383 merged**) ? a deterministic **quality-neutral proof** (freeze+compare the DECISION output = segments + stitch plan off a cached trajectory; **NOT** a reel-MD5). **Decisions D1?D11 LOCKED** (D10 = double-run determinism guard, deterministic segmenters only; D11 = advisory?gating staging). **? NEXT = \$7 P1** (canonical snapshot schema/serializer) ? P2 (`build\_stitch\_plan()` extract) ? P3 (extend `test\_golden\_fixtures` + guard). B1/#385 makes `motion` snapshot-eligible.

22 - **Review-automation patterns established** ([[review-signoff-policy]] + [[review-trust-ci-over-rerun]]): spin off **multi-agent review Workflows** (Ultracode; agents READ-ONLY) ? adversarially verify ? post synthesized findings; **sign-off policy** = LGTM pure-refactors on green (AST-identity vs master), HOLD logic/CI-infra for owner; **written LGTM only** (gh=`avidullu` = owner's own account ? GitHub blocks self-Approve); **trust green CI + a PR `## Verification` section, don't re-run identical CI checks** ? reserve local effort for what CI can't cover (AST-identity, merge-safety, env-noise-vs-real-defect, findings-actually-fixed).

23

24 **? Likely next pick-up:** start the Decision-Snapshot project at **\$7 P1**, or continue the audit (D1 split-main.py / D13 ByteTrack). Serving/alpha remains owner-gated (now needs the Cloud-Run move since the droplet is gone).

25

26 ---

27

28 ### Earlier ? serving/alpha track (2026-06-22/23, UNCHANGED + owner-gated; note the droplet alpha is now retired per above)

29

30 ## ? You are here (2026-06-22)

31 ***(2026-06-23 #342)** engine `origin/master` = **`688d28f`** ? merged a small compute-path fix: `GoogleCloudRunJobClient.run\_job` now guards an unset/empty `GOOGLE\_CLOUD\_PROJECT` (was passing `Optional[str]` into `JobsClient.job\_path(project: str,?)` ? latent mypy `arg-type` seen only with `google-cloud-run` installed + a bogus `projects/None/?` path at runtime). Guard runs before the lazy SDK import (CI-testable, SDK-absent) and raises a clear `RuntimeError`; +`test\_real\_client\_rejects\_missing\_project`. Full suite 1488?, cov 85.86%, all 10 CI green, merged on green per [[working-agreement-merging]]. Follow-up doc [#345](https://github.com/Khelsutra/badminton-highlight-indexer/pull/345) (`9329473`) added the operator note to `deploy/cloudrun/README.md` Gotchas. (Originated as a spawned chip; the live cloud-toggle work above is the sibling track ? see [[current-state]] top.)

32

33 ***(2026-06-22 M11+M12)** engine `origin/master` = **`819e395`** (past #336 M11 + #338 M12; also sibling #333 M6?/api/process, #334 Cloud-Run env, #6d3d054 SPA compute-picker) · khelsutra `origin/master` = `ba38f15` (past #96 Privacy Policy, #100). **The two Claude-doable serving-code items are DONE** (M11 flywheel skeleton #336 + M12 gdrive-api #338 ? both default-OFF/inert, adversarially reviewed, merged on green). **Claude's serving + alpha-shareable code is now complete.** **??? THE ONLY REMAINING ALPHA STEP IS OWNER-DRIVEN: pair on the go-live** (cloudflared tunnel + CF Pages + CF Access + Gemini-key rotation per `khelsutra/deploy/DEPLOY\_ALPHA.md`) ? Claude guides, owner drives the CF dashboard/box. M11 going *live* + M9-consent need the counsel ToS/DPA inputs (the 7-pt checklist in [[ui-driven-cloud-serving-gap]] / this session). Other future engine code: M7 real cloud-serving e2e (owner GCP spend), per-video workspace P1?P6, the rally-quality track (corpus-gated).

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35 ***(2026-06-22 sweep)** engine `origin/master` = **`9b37269`** (+ today's #316/#319/#320/#321/#322/#323) · engine **~1428** tests green (combined-master verified w/ corpus manifest) · **0 open engine PRs** · no GCP VMs. This was a cross-track maintenance sweep (worker fix, nightly scheduler, GCS-data doc, gate-A litmus re-ground, golden-fixtures backlog) ? the serving/alpha forward-plan below is unchanged. ? a sibling's redundant gate-A branch `fix/served-gate-a-regrounded-15clip` sits in the MAIN checkout; concurrency being trimmed.

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37 ***(2026-06-21 prior)** engine `origin/master` = `0c0e064`+ (past #292/#293/#294) · khelsutra `origin/master` = `4627830` (past #63/#64/#65) · **no GCP VMs (audited \$0)** · engine ~1259 tests green.

38 **??? ALPHA-SHAREABLE SESSION (2026-06-21) ? 3 PRs MERGED; Claude's half of the owner's top priority is DONE:**

39 - ****KEY CORRECTION** (anti-spiral [[lesson-verify-engine-surface-before-scoping]]):****** the handoff said "do the B-P2 SPA ingest screen + M7 Cloud Run *service*" ? but ****B-P2** was already shipped****** (khelsutra #56, live e2e in `DRY\_RUN\_CUJ.md` §9) and recon showed alpha-shareable is a ****DEPLOY/AUTH**** task, not an SPA build. CORS is already env-driven; the engine needed no CORS code.

40 - ****Owner decisions** (this session):****** (1) pursue alpha-shareable via topology ******(i) Tunnel-to-a-box******; (2) sub-topology = ****SPLIT**** (SPA on CF ****Pages**** + engine on `api.` ****cloudflared tunnel**** + CF Access over both ? engine `HOSTING\_PLAN.md` §2), single-origin Caddy kept as the alternative.

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55 - [Batteries-included app plan](packaging-app-plan.md) ? downloadable Python app (local web service for inference/training/annotation). ****11 PRs merged 2026-06-21/22**** ? LIGHT-tier bundle-ready (`pip install`?`indexer --app`?bundled SPA, torch-free). Doc `docs/PACKAGING\_PLAN.md` §7. ****01+02 LOCKED**** (cross-platform, LIGHT-first WASB-complete north star); 03+ gated. Pairs [[ui-driven-cloud-serving-gap]]. Don't relitigate locked calls.

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57 ## Monetization (hosted API)

58 - [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM; licensing & per-video cost are first-class; user is Bangalore-based

59 - [Monetization research status](monetization-research-status.md) ? audit done: Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full report in docs/

60 - [Machine-pooling idea](machine-pooling-idea.md) ? owner "food for thought" (2026-06-20): pool GPU machines (queue-on-one-host vs Cloud-Run scale-to-zero) instead of spin-up/kill per task ? born from a single-GPU-slot collision between two sessions (quota=1). Infra-track design item.

61 - [UI-driven cloud-serving gap](ui-driven-cloud-serving-gap.md) ? ****owner directive** (2026-06-21): cloud highlight flow must be UI-driven + ALPHA-SHAREABLE, not SSH/CLI; surface compute options (local vs cloud) in UI at start.****** ? The compute-picker + real Cloud Run L4 GPU e2e validation LANDED 2026-06-22 ? see [[cloudrun-gpu-serving-gotchas]].

62 - [Cloud Run GPU serving gotchas](cloudrun-gpu-serving-gotchas.md) ? ****reusable knobs** to run the WASB worker on Cloud Run L4 GPU****** (region?Mumbai · `--no-gpu-zonal-redundancy` · GCS volume-mount weights · PowerShell-not-Bash `/mnt` flags · `skip\_ai\_handoff=true` no-key · digest-pin?`jobs update` · `[storage-gcs]` in image · control plane needs google-cloud-run+ADC). Verified 2026-06-22 (`artifacts/cloudrun\_e2e\_validation/`).

63 - [khelsutra site droplet parity](khelsutra-site-droplet-parity.md) ? ****RETIRED** 2026-06-25.****** DO droplet `claude-do-rally-test` (dual-purpose site mirror + alpha engine) kept drifting from CF ? owner retired it: data archived to GCS+local, droplet deleted (\$0), monitor removed (khelsutra #117). Follow-ups: `api.` CNAME/CF-Access cleanup. Pairs [[compute-stack-gcp-only]] [[ui-driven-cloud-serving-gap]].

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65 ## Forward-looking

66 - [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate GitHub repo ? corrected 2026-06-06)

67 - [Paper co-author aim](paper-coauthor-aim.md) ? ****standing side-aim****: co-author a paper on "local, cheap & efficient rally detection via multiple clever ideas". Keep `docs/QUALITY\_ITERATIONS.md` as the ablation-ready empirical backbone; honest measured results ONLY.

68

69 ## Project: muneem-ai (SEPARATE root from khelsutra-guru)

70 - [muneem-ai workspace](muneem-ai-workspace.md) ? `C:\Users\avidu\Projects\muneem-ai` = 3 repos (****muneem**** Python local-first finance bookkeeper + ****node-text-detection**** Node/Tesseract OCR + ****Email-Mining**** prior art). Per-session focus switch (2026-07-01); persistent root stays khelsutra-guru. muneem CI = ruff(len 100)/mypy/pytest-cov80, vendored `ruff-bin/ruff.exe`. Detail [[muneem-ai-workspace]].

71 - [muneem grocery-replenishment SENSOR](muneem-grocery-replenishment.md) ? ****GR.0?GR.5 BUILT + MERGED to main 2026-07-01 (9 PRs #100?108).**** Loop = `grocery ingest`/`ingest-receipt --store` ? `pantry due`/`pantry compare`. A depletion ****sensor**** (owner: "as a sensor, not a delight"): per-item history (`purchase\_items`, migration v8) ? slow-staple run-out + confidence + cheapest-store; 2nd source via OCR/text receipt. Built on 206 real BigBasket line-items from ****avidullu.svp@**** Gmail. GR.5 = deterministic structural canonicalization; ****the LEARNED Tier-1 is the ONLY unbuilt piece ? parked under §8.5 (owner-gated).**** Detail [[muneem-grocery-

replenishment]].

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OPEN

Hosted alpha: split-origin (Pages app. + tunnel api.) breaks Cloudflare Access for the SPA ? single-origin needed

Symptom

On the hosted alpha, the SPA loaded at `app.khelsutra.guru` but **every** upload failed "Can't reach KhelSutra. Is it running?" ? even though the engine was healthy.

Root cause (NOT file size)

`app.khelsutra.guru` (Cloudflare Pages) and `api.khelsutra.guru` (cloudflared tunnel) are **different origins**. The SPA's `fetch` (`web/src/api/client.ts:60`) sends **no `credentials: 'include'`**, so cross-origin calls to `api.` never carry the Cloudflare Access cookie ? Access `302`s every API call to the login page ? the fetch rejects ? "can't reach." It fails on **any** upload, not just large ones.

Verified it's not size: a real **95 MiB** chunk PUT through the tunnel returned 200 in ~2s (and 50 MiB ? 200).

Fix applied (2026-06-22) ? single-origin

The engine already bundles the SPA (P1b), so I repointed to **single-origin**:

`api.khelsutra.guru` now serves the **SPA** at `/` and the **API** at `/api` behind **one** Access cookie. Verified end-to-end via a service token: `GET /` ? 200 (`<title>KhelSutra Studio</title>`), `/api/health` ? 200, `upload/init` + 95 MiB + 50 MiB chunk PUTs ? 200.

Remaining / to pick up

- app.khelsutra.guru** DNS needs repointing to the tunnel: `CNAME app ? 55142398-732c-4803-8a9c-48766f1f5cac.cfargotunnel.com`, proxied. Blocked on the CF API token lacking `Zone.DNS:Edit` (the scope add didn't take); do via dashboard or fix the token. (The Pages custom domain for `app.` was removed.)
- Update** `deploy/DEPLOY\_ALPHA.md` to recommend **single-origin** (engine-served SPA) as the default for the SPA+Access case, OR document the cross-origin requirements if split is desired: SPA `credentials:'include'` + engine CORS `allow\_credentials=true` + a way to pre-warm the `api.` Access cookie (the hard part for fetch).
- Refresh** the engine's bundled SPA (currently commit #68) to latest, so the localized reel-watermark (LW2 #73) + recent polish (#82/#83/#86) ship in the hosted app.
- Lesson:** a service-token full-chai

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Structured output provided successfully